SketchThinker-R1: Towards Efficient Sketch-Style Reasoning in Large Multimodal Models

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EDIT — Group members & contributions

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[Name A]	[contribution]	[xx%]
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12 min

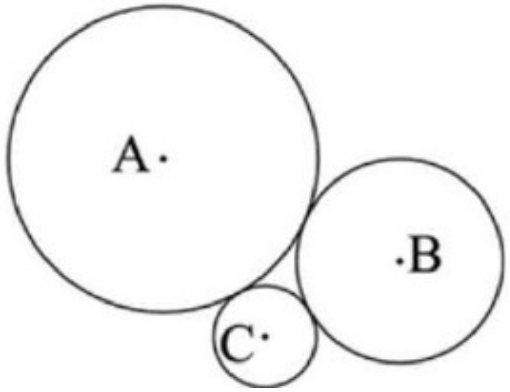
Paper analysis

3 min

Research pitch

Long reasoning spends tokens on repeated explanation

Image + question



Three externally tangent circles

Find the area between centers

Verbose explanation



Textual sketch

Tangency → distances → triangle → area

Goal: retain the key logical flow

PDF pp.1–2, Fig.1; p.8, Fig.3

Three training stages turn sketches into a policy

1 · Cold Start

GPT-5 rewrites long traces

Supervise the answering model

2 · SketchJudge

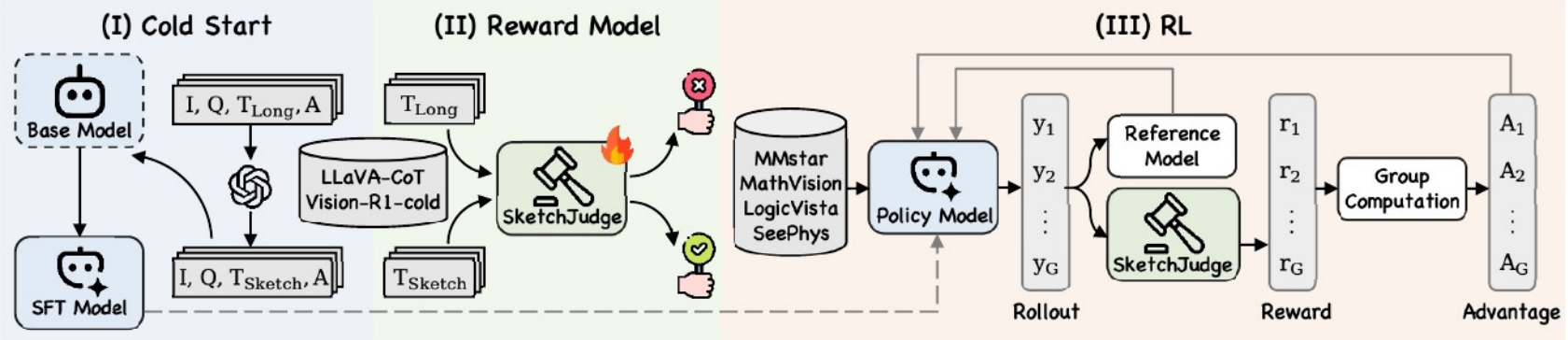
Train long / sketch labels

Score thinking text only

3 · Sketch RL

GRPO on new questions

Accuracy + format + style



Top: redrawn pipeline · Bottom: original Figure 2

Cold start teaches the model what a sketch looks like

Existing long-trace source	Sampled questions
LLaVA-CoT-100K	10,000
Vision-R1-cold	10,000

20K

SketchColdStart-20K

Long trace → GPT-5 rewrite → numbered sketch

Supervised next-token training

Qwen2.5-VL-7B-Instruct · also evaluated at 3B

PDF p.5, §3; p.14, App.D.1; p.21, Table16

SketchJudge rewards a textual style

Original long trace Label 0

Rewritten sketch Label 1

Qwen2.5-7B-Instruct

40K labeled thinking traces

Binary style judgment

Judge input: thinking text only

Sees neither the image nor the question

Style label ≠ verified visual support

RL balances correctness, format, and thinking style

R = 0.5 accuracy + 0.4 format + 0.1 thinking-style

Illustrative response	Accuracy	Format	Style	Reward
Correct, valid format, sketch	1	1	1	1.0
Wrong, valid format, sketch	0	1	1	0.5

Five rollouts → relative advantages → clipped policy update

KL penalty anchors the policy · no default token penalty

PDF p.4, Eqs.2–6; p.5, §3

Evaluate answer quality and reasoning cost separately

Evaluation domain	Benchmark
Broad reasoning	MMMU
Visual mathematics	MATH-Vision
Visual logic	VisuLogic
Visual physics	PhyX

Report separately

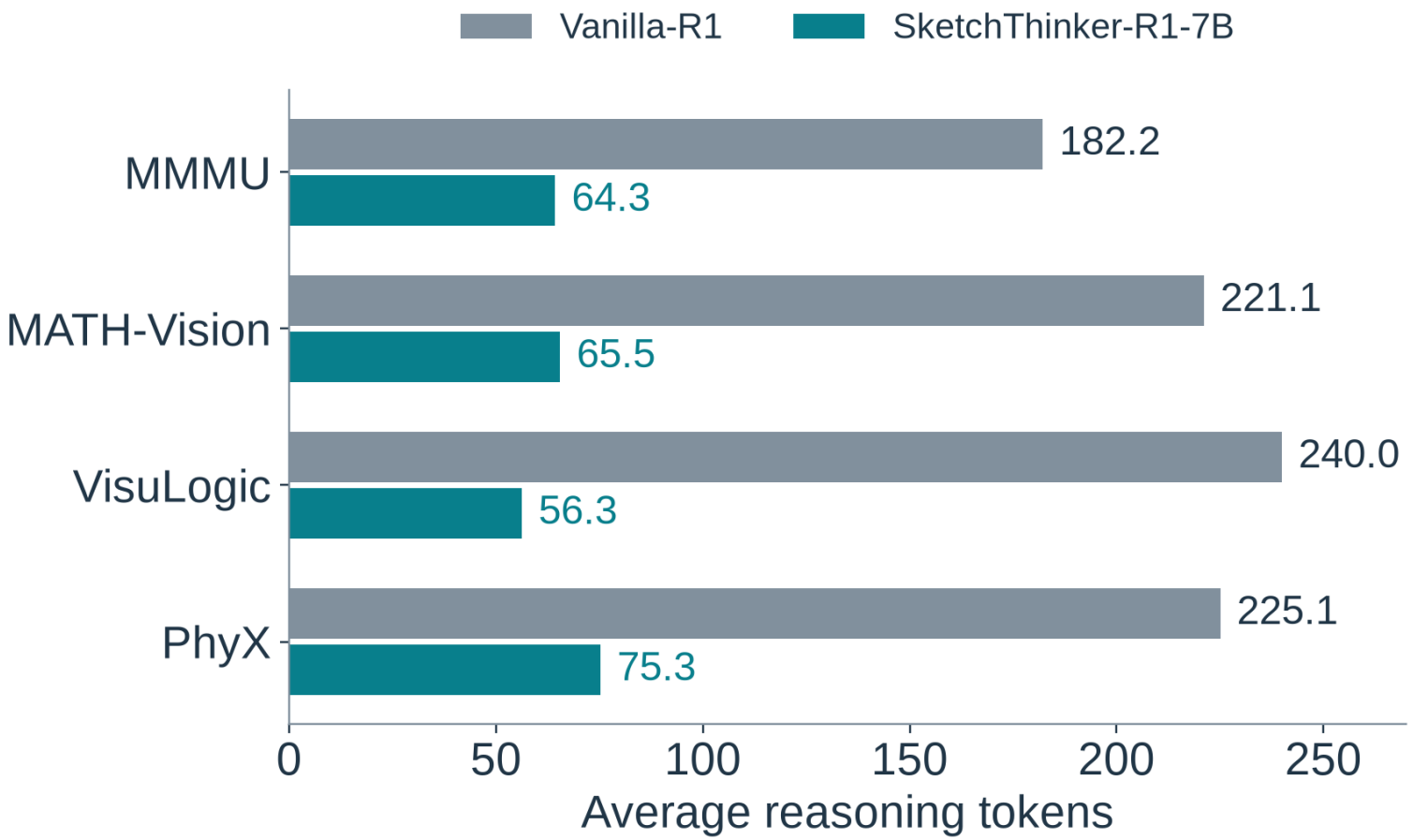
Answer accuracy (%)

Average reasoning tokens

EoT = accuracy / tokens

RL training: 250 questions × 4 sources = SketchRL-1K

At 7B, tokens fall sharply across all four benchmarks

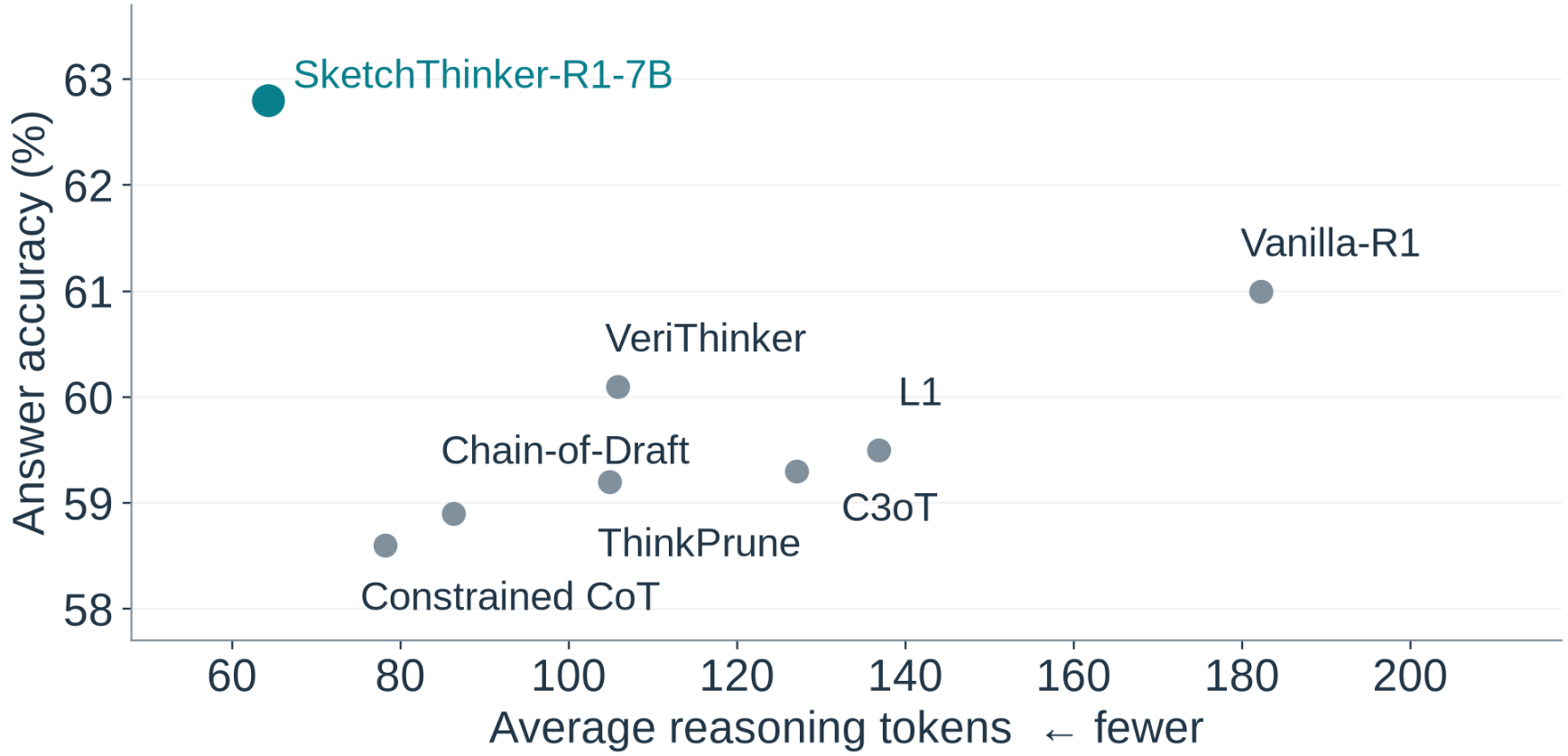


Accuracy (%)	Vanilla	Sketch
MMMU	61.0	62.8
MATH-Vision	31.0	31.7
VisuLogic	27.6	27.8
PhyX	46.7	48.6

64.7–76.5% fewer tokens

PDF p.6, Table1; reductions computed from table entries

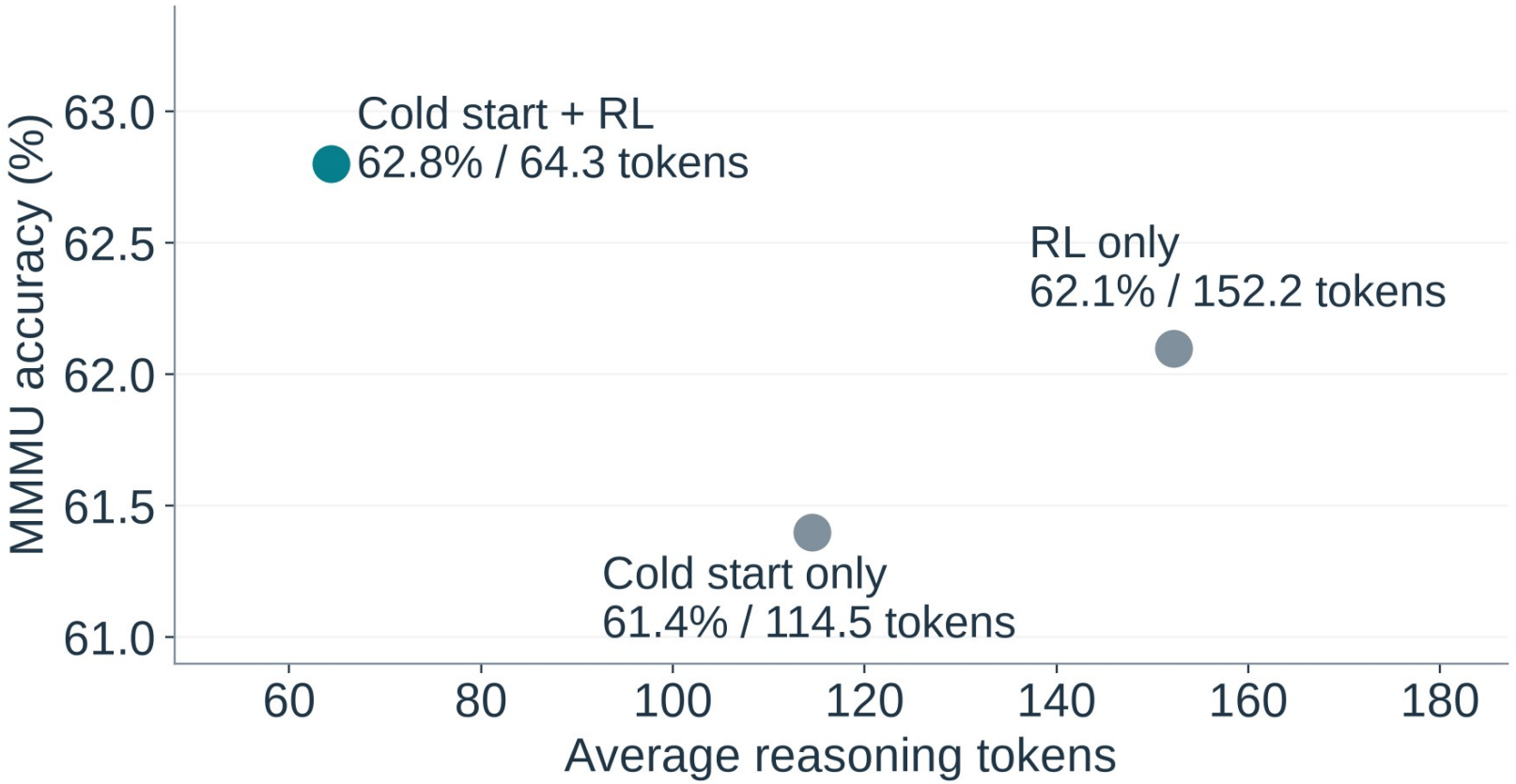
On MMMU, the 7B model improves both axes



Reported operating points · uncertainty not reported

PDF p.6, Table1, MMMU columns

Cold start and RL contribute together



Judge fine tuning

With SFT

62.8% / 64.3 tokens

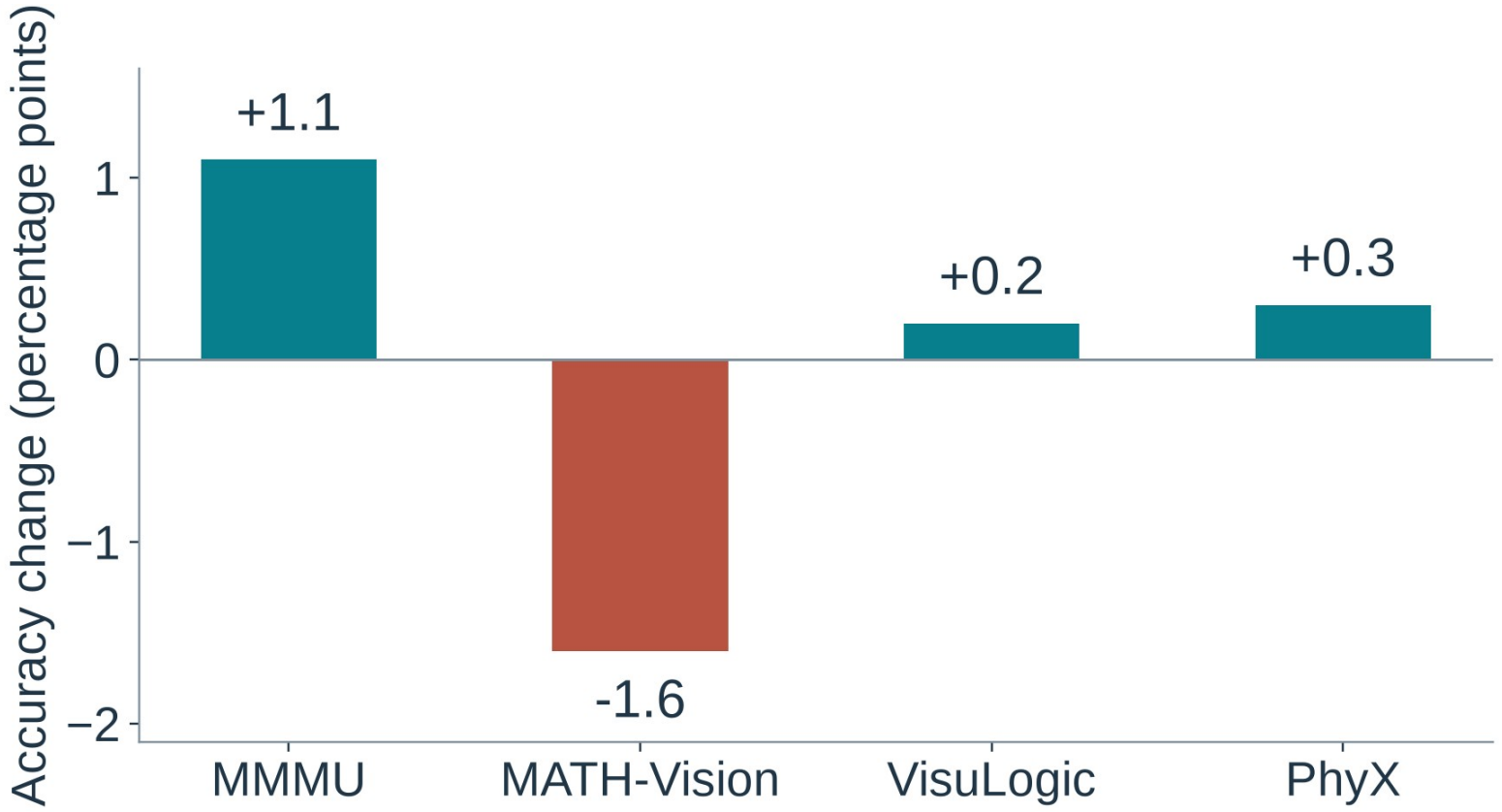
Without SFT

61.0% / 72.1 tokens

MMMU · 7B backbone

PDF p.7, Tables3a,3b,3d

The 3B result exposes an accuracy boundary



MATH-Vision

26.9% → 25.3%

Reasoning tokens

151.2 → 72.7

Token savings do not guarantee accuracy gains

PDF p.6, Table2

Appendix results refine the efficiency claim

MMMU reward schedule	Accuracy (%)	Tokens
Fixed	62.8	64.3
Staged	63.0	63.8
Dynamic	63.2	62.5

Training time

2.70 h → 2.21 h · 8 H800 GPUs

More RL data

1K → 5K samples

62.8% → 66.1% MMMU

Human readability study

20 jointly correct examples

5 evaluators

PDF pp.16–18, Tables7,9,11; p.20, Table15

Efficient answers leave evidence preservation unresolved

What the paper establishes

Large token savings in the tested 7B comparisons

Grounding	Text-only style judge
Uncertainty	Point estimates; small human audit
Reproduction	Rollout batch size: 128 vs 512

Open question: what evidence survives compression?

Pitch: preserve the evidence behind an answer

Preserving Visual Evidence in Compressed Multimodal Reasoning

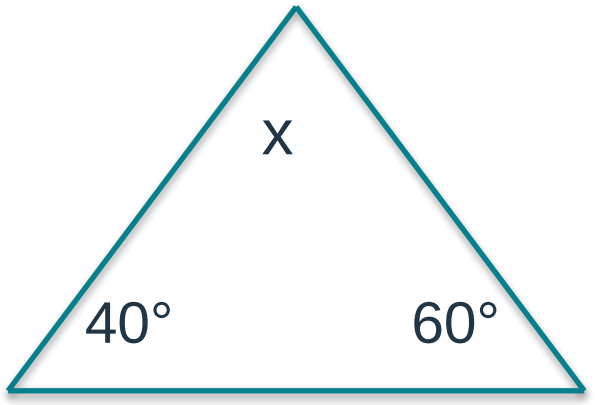


Image-only angle readings

Sufficient sketch

Angles read: 40° and 60°

Triangle sum: x = 180° − 40° − 60° = 80°

Explicit premises + valid deductions

Accept alternative complete proofs

Compare evidence and style under shared data and compute

Shared cold start + rewritten data	Auxiliary reward
Evidence arm	Image + question + trace
Style arm	Thinking text only
Zero-component arm	No auxiliary component

2,400 rendered problems · checked image pairs

Match achieved tokens within 5% · include judge compute

Measure: accuracy · premise retention · human sufficiency · paired correctness

Proposal §§3–4; all sample sizes and targets are planned

A six-month plan with explicit success and null outcomes

M1	M2	M3	M4	M5	M6
Reproduce	Annotate	Calibrate	Train	Controls	Human audit
Compute pilot	Checked pairs	Judge gate	Three seeds	Frozen tests	Release

Judge gate: false acceptance ≤10% · sufficient recall ≥70%

Success: higher retention and accuracy at matched resources

Null results and failure cases are deliverables